Towards a Secure and Seamless Digital Energy Future for Viksit Bharat

(India Energy Stack, Data Exchange & Privacy and Cyber Resilience)

Distribution Utility Meet 2026

Power Foundation of India

28th Oct, 2026
New Delhi

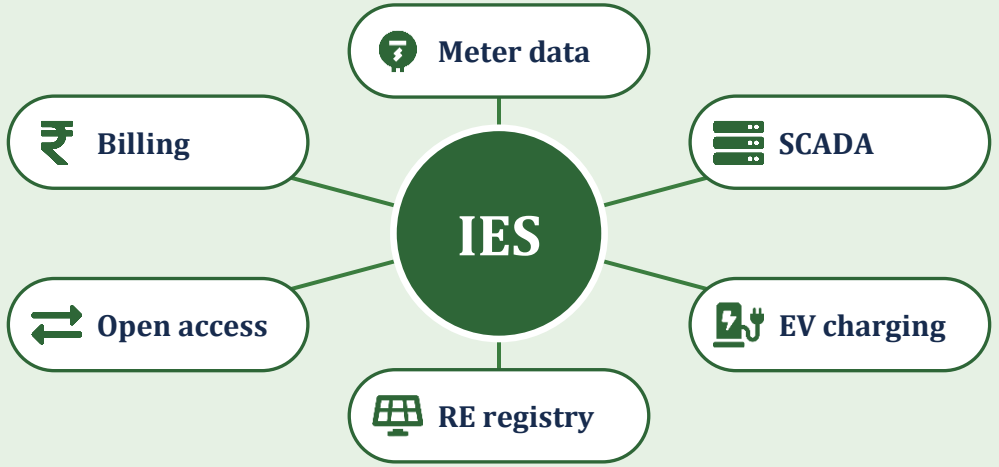
The grid is going digital

More devices. More data. More risk.

TODAY: DATA SILOS

- Billing
- Meter data
- SCADA
- Open access
- RE registry
- EV charging

***TOMORROW: CONNECTED**



DEVICES

19.58 crore

smart meters sanctioned (RDSS)

DATA

66

datasets identified for sharing

RISK

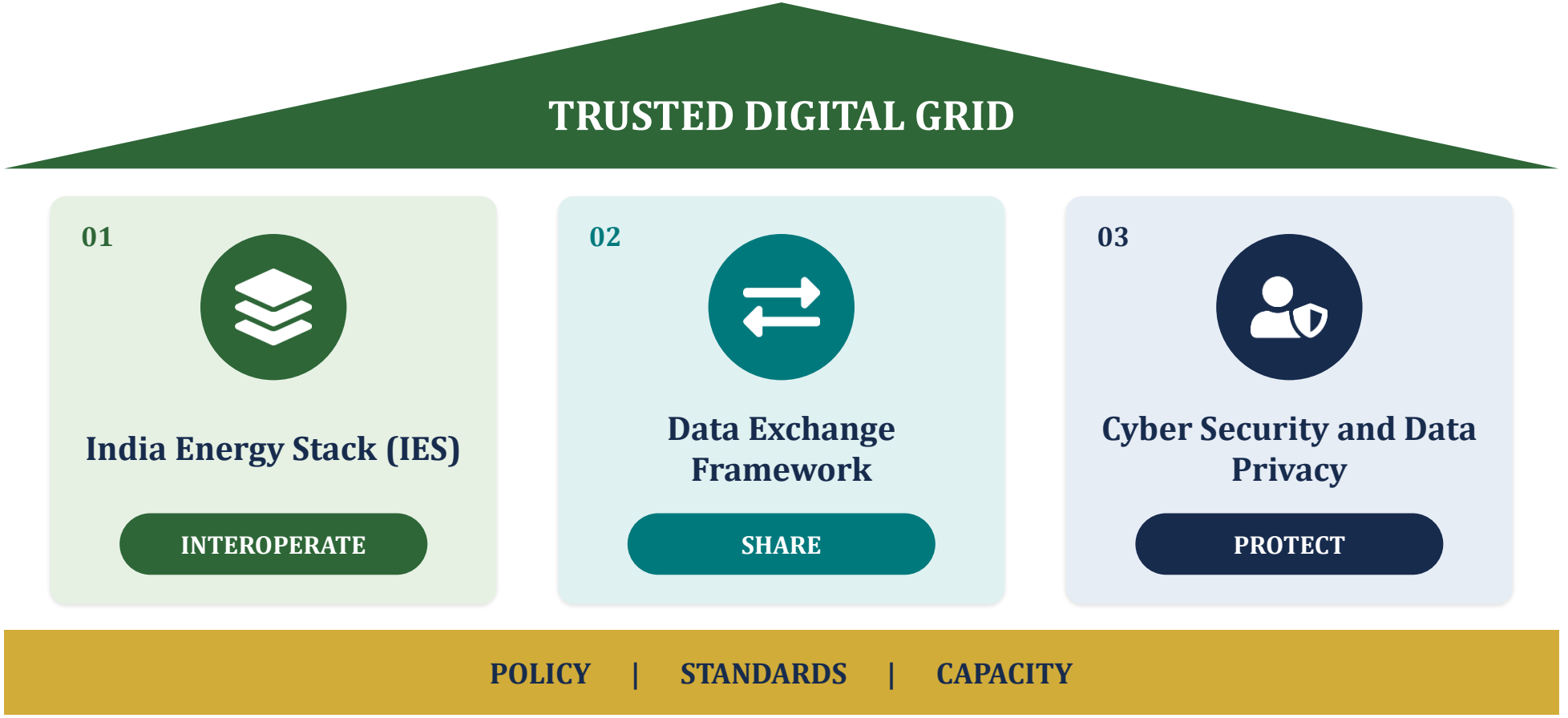
6 hours

to report a cyber incident

Source: RDSS dashboard; MoP Draft Electricity Data Sharing Framework, 2026; CEA Cyber Security Regulations, 2026

One trust architecture for the digital grid

Three building blocks. One outcome.



India Energy Stack: DPI for the power sector

Aadhaar did it for identity, UPI for payments. IES aims to do it for energy.

Aadhaar
Identity

UPI
Payments

IES
Energy



OWNER	NODAL AGENCY	KNOWLEDGE PARTNER	TASKFORCE CHAIR
Ministry of Power	REC Limited	FSR Global	Dr. R.S. Sharma

Source: Ministry of Power; REC Ltd; IES Strategy and Architecture documents, v0.4 (Mar 2026)

IES journey and the first live trade

From taskforce to a live peer-to-peer trade in eight months.

Jun 2025 — Taskforce constituted

Nov 2025 — Strategy and architecture v0.1

Dec 2025 — v0.2 and P2P pilot specs

Feb 2026 — v0.3 and first live P2P trade

Mar 2026 — v0.4 documents released

2026-27 — DISCOM pilots, then scale

FIRST LIVE P2P TRADE ON IES | India AI Impact Summit, February 2026



6 units traded

₹30 earned by farmer

Inter-state: UP to Delhi

Source: MoP and REC press releases (Nov 2025 to Mar 2026)

IES use cases under test

Energy Credentials

- No common credential standard
- **Regulator-notified schema**

P2P Energy Exchange

- No regulated product yet
- **Wheeling + settlement rules**

Tariff Intelligence

- Tariffs locked in PDFs
- **SERC-published structured data**

Regulatory Data Exchange

- Manual, template reporting
- **API filing, mandated formats**

Energy Digest (Meter Data)

- Meter rollout, consent UX
- **Consent manager + data standards**

DER Integration

- No DISCOM visibility of DERs
- **Standard registry + telemetry**

Maturity (indicative): Concept > Pilot > Scale-ready > Regulated

Data Exchange Framework (Draft 2026)

National Electricity Data Sharing Framework, issued by MoP on 22 June 2026.

DATA OWNERS

- Generators
- Transmission
- DISCOMs
- Load despatch centres
- Regulators

NEDC + NEDP

National Electricity Data Centre and Portal

- Metadata catalogue
- Standard formats by CEA
- APIs and secure data rooms

DATA USERS

- Researchers
- AI start-ups
- EV and urban planners
- Green finance
- Policy makers

66
datasets identified

12 months
to publish metadata catalogue

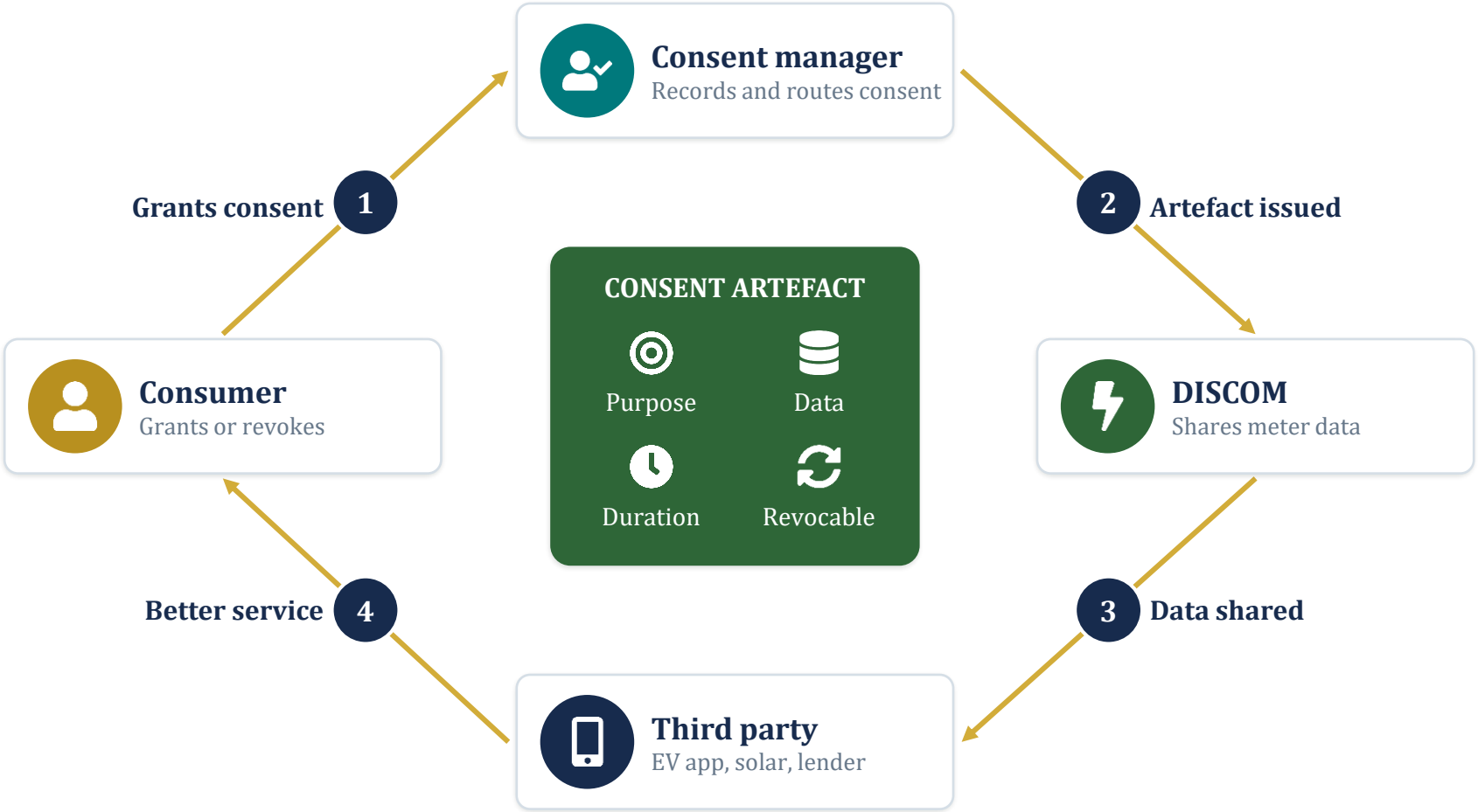
18 months
to be searchable on NEDC

Voluntary
ownership stays with the source

Source: MoP, Draft National Electricity Data Sharing Framework, 22 June 2026

Consent-led sharing of consumer data

The consumer decides who sees their data, why, and for how long.



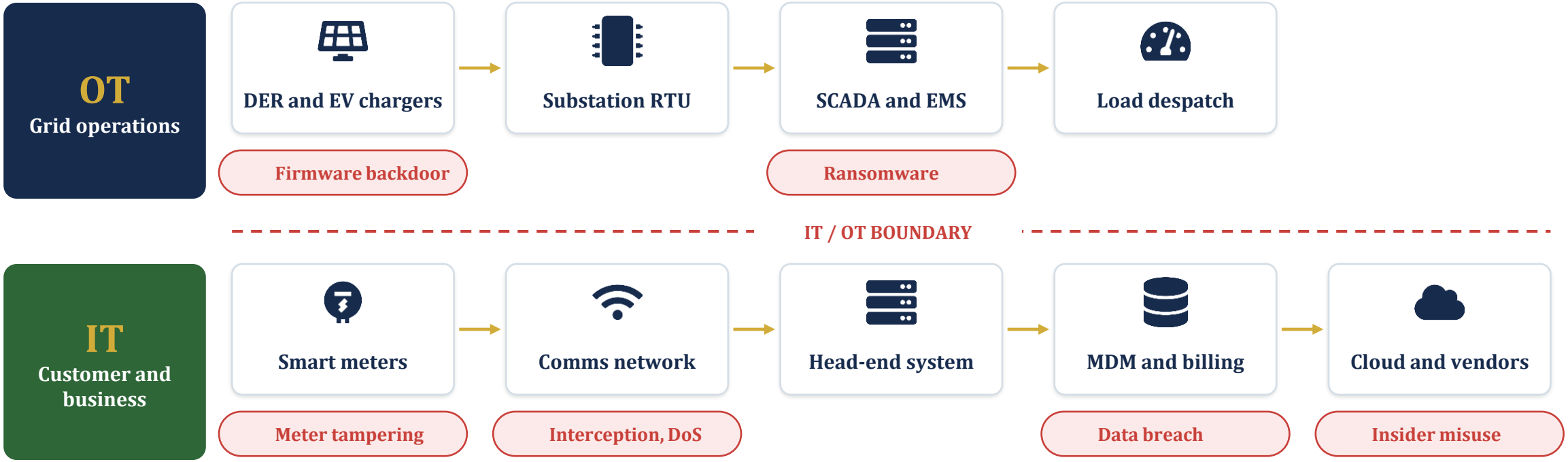
Revoke anytime
As easy as giving consent

Audit trail
Every access logged

DPDP aligned
Consent Managers live from 13 Nov 2026

Digital grid: an expanding attack surface

Every connected device is a potential entry point.



Security must be designed in, not bolted on.

CEA Cyber Security Regulations, 2026

Notified 31 July 2026. Binding from 1 April 2027. Replaces the 2021 guidelines.

KEY TIMELINES

6 hrs
Report incidents to CSIRT-Power, CERT-In

24 hrs
Report cyber sabotage

1 month
Fix critical and high audit findings

3 months
Fix medium and low findings

9 to 15
Months between annual audits

3 yrs
Minimum CISO tenure

KEY MANDATES

- **CISO and 24x7 security division in India**
- **OT isolated from IT and internet**
- **Trusted supply chain with SBOM**
- **Encrypted data stored in India**

INCIDENT REPORTING CHAIN

Utility
CISO and 24x7 ISD

→ 6 hrs →

CSIRT-Power
At CEA, since 2023

↔

CERT-In
National agency

6 SECTORAL CERTs

- Thermal
- Hydro
- Transmission
- Grid operation
- Renewable
- Distribution

Source: CEA (Cyber Security in Power Sector) Regulations, 2026; PIB, Dec 2025

Data privacy: DPDP readiness for DISCOMs

Every DISCOM is a data fiduciary for millions of consumers.

13 Nov 2025
Rules notified, Data Protection Board

13 Nov 2026
Consent Managers go live

13 May 2027
Full obligations apply

We are here

Notice and consent

Clear, specific, revocable

Security safeguards

Encryption, masking, logs kept 1 year

Breach reporting

Detailed report to the Board in 72 hrs

Consumer rights

Access, correct, erase

Retention limits

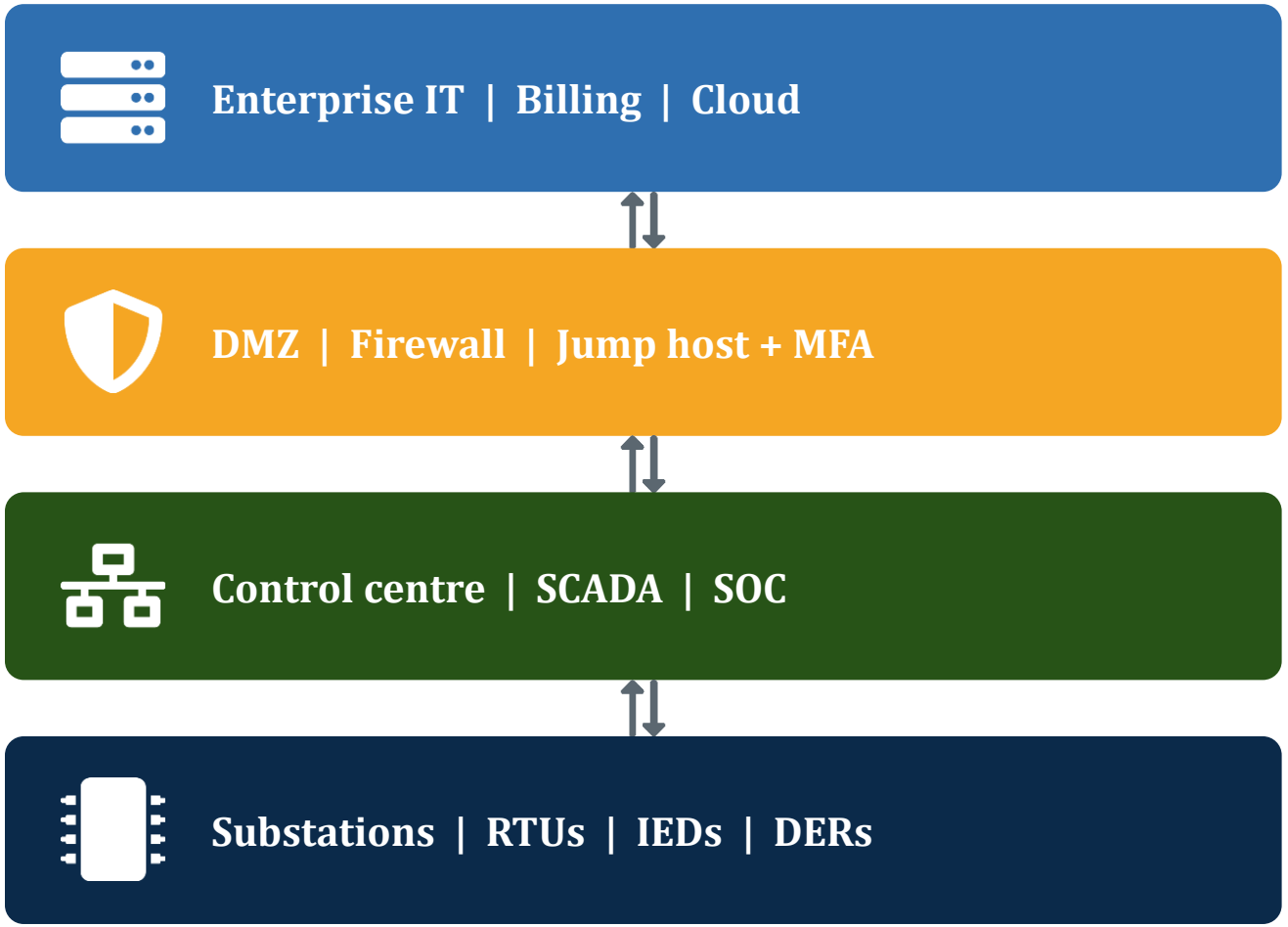
Erase when purpose ends

Penalty up to **₹250 crore** for failing to safeguard personal data

Source: Digital Personal Data Protection Act, 2023; DPDP Rules, 2025

OT security for legacy and modern grid assets

IEC 62443 and NIST SP 800-82 in practice



IEC 62443

- Zones and conduits
- Security levels SL1-4
- Vendor secure lifecycle

NIST SP 800-82

- ICS risk guidance
- Architecture and controls
- Passive OT monitoring

Securing the supply chain and product trust

Build trust before the product is energized, not after an incident

Design	Build	Ship	Install	Operate
Hidden backdoors	Opaque components	Tampered firmware	Unverified product	Vendor remote access
Trusted-source rules	SBOM in every tender	Signed firmware	Lab + acceptance tests	Time-bound, audited access

From paper CCMPs to scenario-based drills

A plan that has never been exercised is only a hypothesis



Scenarios to rehearse

- Ransomware on billing
- SCADA compromise
- Rogue DER / meter commands
- Poisoned vendor firmware

2x / year	1x / year	Joint
Table-top	Technical	Grid-level

What compliance costs and how it gets funded

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RDSS, ARR/tariff or a ring-fenced opex line?

Where the money goes

- 24x7 monitoring / SOC
- Segmentation + tools
- Scarce OT skills
- Audits + drills

Who pays?

RDSS

- ✓ Central capex support
- ✗ Cyber eligibility unclear

ARR / Tariff

- ✓ Sustainable and auditable
- ✗ SERC must accept as prudent cost

Ring-fenced opex

- ✓ Budget protected, trackable
- ✗ Needs board + regulator buy-in

Likely answer: a blend, with a regulator-approved cyber cost template

The way forward: a roadmap for DISCOMs

Prepare now. Comply. Scale beyond.

PHASE 1 PREPARE

- Appoint CISO and Data Governance Officer
- Map and classify data by tier
- Plan IT/OT segregation
- DPDP gap assessment

PHASE 2 COMPLY

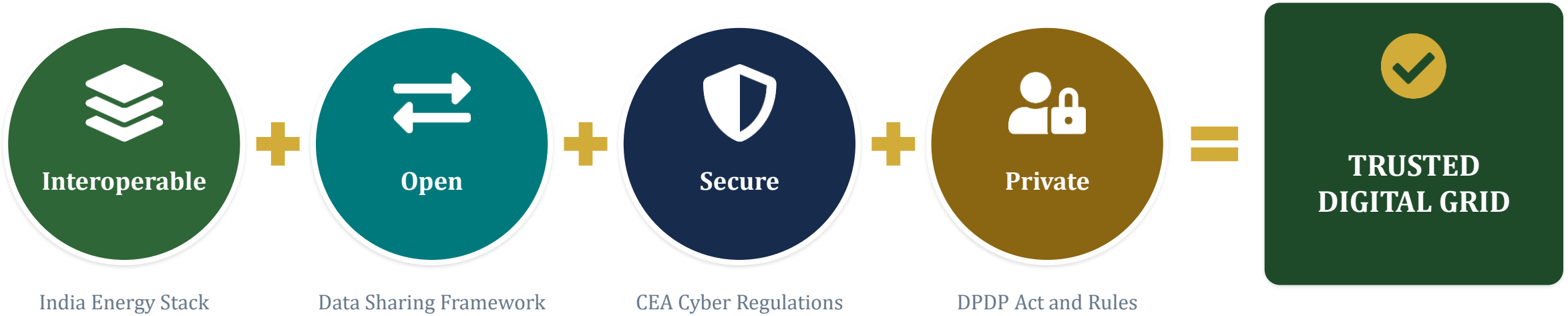
- 24x7 security operations
- Consent and notice systems live
- VAPT, SBOM and annual audits
- Publish Tier 1 data on NEDP

PHASE 3 SCALE

- Join IES pilots and use cases
- Tier 2 data via secure data rooms
- Continuous monitoring and drills
- Skill teams and vendors

Building the trusted digital grid

Four properties. One outcome.



India Energy Stack

Data Sharing Framework

CEA Cyber Regulations

DPDP Act and Rules

Data creates value only when it is trusted.

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Thank You